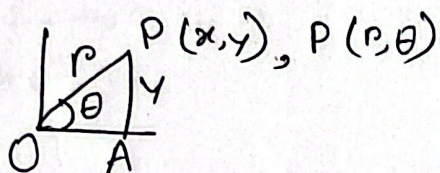


Relation betⁿ C & P (Cartesian, Polar)



ch-2

Area, $OA = x$

$AP = y$

$OP = r$

$$\Rightarrow OP^2 = OA^2 + AP^2$$

$$\Rightarrow r^2 = x^2 + y^2$$

$$\boxed{r = \sqrt{x^2 + y^2}}$$

Now, acute $\cos \theta = \frac{OA}{OP}$

$$\Rightarrow \cos \theta = \frac{x}{r}$$

$$\boxed{x = r \cos \theta} \rightarrow \textcircled{1}$$

$$\text{, } \sin \theta = \frac{AP}{OP}$$

$$\Rightarrow \sin \theta = \frac{y}{r}$$

$$\boxed{y = r \sin \theta} \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$

$$\frac{y}{x} = \frac{r \sin \theta}{r \cos \theta}$$

$$\Rightarrow \frac{y}{x} = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1} \frac{y}{x}$$

C to P

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \frac{y}{x}$$

P to C

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Distance

$$\Rightarrow P_1 P_2 = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 (\cos \theta_2 - \cos \theta_1)}$$

Area. Tri.

$$\Rightarrow A = \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) - \frac{1}{2} r_3 r_1 \sin(\theta_1 - \theta_3)$$

Cartesian to Polar

Example 5) i)

$$9x^2 + 4y^2 = 36$$

$$\Rightarrow 9r^2 \cos^2 \theta + 4r^2 \sin^2 \theta = 36$$

$$\Rightarrow r^2 (9 \cos^2 \theta + 4 \sin^2 \theta) = 36$$

$$\Rightarrow r^2 (5 \cos^2 \theta + 4 \cos^2 \theta + 4 \sin^2 \theta) = 36$$

$$\Rightarrow r^2 (5 \cos^2 \theta + 4 (\cos^2 \theta + \sin^2 \theta)) = 36$$

$$\Rightarrow r^2 (4 + 5 \cos^2 \theta) = 36$$

$$9x^2 + 4y^2 = 36$$

$$\Rightarrow 5x^2 + 4x^2 + 4y^2 = 36$$

$$\Rightarrow 5x^2 + 4(x^2 + y^2) = 36$$

$$\Rightarrow 5x^2 + 4r^2 = 36$$

$$\Rightarrow 5r^2 \cos^2 \theta + 4r^2 = 36$$

$$\Rightarrow r^2 (4 + 5 \cos^2 \theta) = 36$$

Exercise 5) ii)

$$x^4 + x^2 y^2 - (x+y)^2 = 0$$

$$\Rightarrow r^4 \cos^4 \theta + r^2 \cos^2 \theta \cdot r^2 \sin^2 \theta - (r \cos \theta + r \sin \theta)^2 = 0$$

$$\Rightarrow r^4 \cos^4 \theta + r^4 \cos^2 \theta \sin^2 \theta - (r^2 \cos^2 \theta + 2r^2 \cos \theta \sin \theta + r^2 \sin^2 \theta) = 0$$

$$\Rightarrow r^4 \cos^4 \theta + r^4 \cos^2 \theta \sin^2 \theta - r^2 \cos^2 \theta - 2r^2 \cos \theta \sin \theta - r^2 \sin^2 \theta = 0$$

$$\Rightarrow r^4 \cos^4 \theta + r^4 \cos^2 \theta \sin^2 \theta - 2r^2 \cos \theta \sin \theta - r^2 (\cos^2 \theta + \sin^2 \theta) = 0$$

$$\Rightarrow r^2 \cos^4 \theta + r^2 \cos^2 \theta \sin^2 \theta - 2 \cos \theta \sin \theta - 1 = 0$$

$$\Rightarrow r^2 \cos^4 \theta + r^2 \cos^2 \theta \sin^2 \theta - \sin^2 \theta - 1 = 0$$

$$\Rightarrow x^2 (x^2 + y^2) - (x+y)^2 = 0$$

$$\Rightarrow x^2 (x^2 + y^2) - x^2 - 2xy - y^2 = 0$$

$$\Rightarrow x^2 (r^2) - 2xy - (x^2 + y^2) = 0$$

$$\Rightarrow x^2 (r^2) - 2xy - r^2 = 0$$

$$\Rightarrow r^4 \cos^2 \theta - 2r^2 \cos \theta \sin \theta - r^2 = 0$$

$$\Rightarrow r^2 (r^2 \cos^2 \theta - 2 \cos \theta \sin \theta - 1) = 0$$

$$\Rightarrow r^2 \cos^2 \theta = 1 + 2 \cos \theta \sin \theta$$

$$\Rightarrow r^2 = \frac{1}{\cos^2 \theta} + \frac{2 \cos \theta \sin \theta}{\cos^2 \theta}$$

$$\Rightarrow r^2 = \sec^2 \theta + 2 \tan \theta$$

$$\Rightarrow r^2 = 1 + \tan^2 \theta + 2 \tan \theta$$

$$\Rightarrow r^2 = (1 + \tan \theta)^2$$

$$\Rightarrow r = \pm (1 + \tan \theta)$$

cglara chr-1

Pg-10

1/8/2

Ex: 6, 3, 4, 5
Example
Exercises 4, 5, 6

* Transform to Cartesian co-ordinates

$$r(\cos 3\theta + \sin 3\theta) = 5k \sin \theta \cos \theta$$

$$\Rightarrow r(4\cos^3 \theta - 3\cos \theta + 3\sin \theta - 4\sin^3 \theta) = 5k \sin \theta \cos \theta$$

$$\Rightarrow r(4(\frac{x}{r})^3 - 3\frac{x}{r} + 3\frac{y}{r} - 4(\frac{y}{r})^3) = 5k \frac{x}{r} \frac{y}{r}$$

$$\Rightarrow \frac{4x^3}{r^2} - 3x + 3y - \frac{4y^3}{r^2} = 5k \frac{xy}{r^2}$$

$$\Rightarrow \frac{4x^3 - 4y^3}{r^2} - 3x + 3y = 5k \frac{xy}{r^2}$$

$$\Rightarrow 4(x^3 - y^3) - 3(x - y)r^2 = 5kxy$$

$$\Rightarrow 4(x^3 - y^3) - 3(x^2 + y^2)(x - y) = 5kxy$$

$$r^2 - 2r(\cos \theta - \sin \theta) - 7 = 0$$

$$\Rightarrow r^2 - 2r(\frac{x}{r} - \frac{y}{r}) - 7 = 0$$

$$\Rightarrow r^2 - 2(x - y) - 7 = 0$$

$$\Rightarrow (x^2 + y^2) - 2(x - y) - 7 = 0$$

$$(x^2 + y^2) \pm 2x \mp 7 = 0$$



Exercise:

4) i) $(r_1, \theta_1), (r_2, \theta_2), (r_3, \theta_3)$
 $(-3, 30^\circ), (5, 150^\circ), (7, 210^\circ)$

$$\Delta = \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) + \frac{1}{2} r_3 r_1 \sin(\theta_1 - \theta_3)$$

$$= \frac{1}{2} (-3 \cdot 5) \sin(150 - 30) + \frac{1}{2} (5 \cdot 7) \sin(210 - 150) + \frac{1}{2} (-3 \cdot 7) \sin(30 - 210)$$

$$= -7.5 \sin(120) + 17.5 \sin(60) - 10.5 \sin(-180)$$

(The - inside only shows direction, use + in calculator)

$$\Delta = \frac{7\sqrt{3}}{2} \quad \Delta = 5\sqrt{3}$$

4) ii) $(a, \theta), (2a, \theta + \pi/3), (3a, \theta + 2\pi/3)$

$$\Delta = \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) + \frac{1}{2} r_3 r_1 \sin(\theta_1 - \theta_3)$$

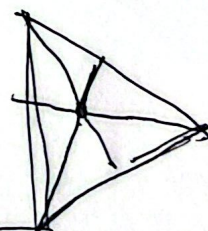
$$= \frac{1}{2} (a \cdot 2a) \sin(\theta + \pi/3 - \theta) + \frac{1}{2} (2a \cdot 3a) \sin(\theta + 2\pi/3 - \theta - \pi/3) + \frac{1}{2} (a \cdot 3a) \sin(\theta - \theta - 2\pi/3)$$

$$= a^2 \sin(\pi/3) + 3a^2 \sin(\pi/3) + \frac{3}{2} a^2 \sin(-2\pi/3)$$

$$= a^2 \left\{ \sin(60) + 3 \sin(60) + \frac{3}{2} \sin(-120) \right\}$$

(The - inside only shows direction, put + in calculator)

$$= \frac{5\sqrt{3}}{4} a^2 - \frac{3\sqrt{3}}{4} a^2 = \frac{2\sqrt{3}}{4} a^2 = \frac{\sqrt{3}}{2} a^2$$



$$\Delta = \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) + \frac{1}{2} r_3 r_1 \sin(\theta_1 - \theta_3)$$

$$= \frac{1}{2} r_1 r_2 \sin(\theta_2 - \theta_1) + \frac{1}{2} r_2 r_3 \sin(\theta_3 - \theta_2) - \frac{1}{2} r_1 r_3 \sin(\theta_3 - \theta_1)$$

Exercise

5) i) $(x^2+y^2)^2 = 2a^2xy$ $[x^2+y^2 = r^2]$

$\Rightarrow r^4 = 2a^2 r^2 \cos \theta \sin \theta$

$\Rightarrow r^2 = 2 \cos \theta \sin \theta a^2$

$\Rightarrow r^2 = a^2 \sin 2\theta$

Exercise

6) i) $r^2 - 2r(\cos \theta - \sin \theta) - 7 = 0$

$\Rightarrow x^2 + y^2 - 2r(\frac{x}{r} - \frac{y}{r}) - 7 = 0$

$\Rightarrow x^2 + y^2 - 2x + 2y - 7 = 0$

$\Rightarrow x^2 + y^2 - 2x + 2y - 7 = 0$

ii) $r(1 - e \cos \theta) = ep$

$\Rightarrow r - er \cos \theta = ep$

$\Rightarrow r - ep \frac{x}{p} = ep$

$\Rightarrow r = ex + ep$

$\Rightarrow \sqrt{x^2 + y^2} = ex + ep$

$\Rightarrow x^2 + y^2 = (ex + ep)^2$

